SmolVLA — Gesture Mimic Training

Chunk-size vs. Accuracy / Throughput Study

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Setup

Training & Evaluation Data

Combined dataset — 3 sources @ 30 fps (train + held-out test, disjoint)

Dataset (human position)	Train set		Test set (held-out)	
	Episodes	Frames	Episodes	Frames
`multigesture_mimic` — *sitting pose*	39	15,015	10	3,868
`extended_gesture_mimic` — *standing pose*	35	20,787	9	5,346
`Action-per-video-…clean` — *one action / video* (80/20 split, seed 42)	315	75,446	79	18,930
Total	**389**	**111,248**	**98**	**28,144**

Model: SmolVLA | **Training:** `chunk_size=50`, `n_action_steps=10`, bf16, batch 64, lr 5e-6, 30 k steps, RTC on (`exec_horizon=10`) | Dataset 3: 56 bad episodes removed before split.

Sweep variable: parallel actions per inference call $k \in \{1, 2, 4, 6, 8, 10, 12, 14\}$ — controls **throughput** ↔ **accuracy** trade-off.

Accuracy vs. Action-Chunk Size

Mean MAE per test set (lower is better) — checkpoint 030000

Actions / step (k)	multi-gesture	eep it	action-per-episode
1	1.75	2.57	1.99
2	1.83	2.56	2.15
4	1.92	2.76	2.32
6	2.02	2.93	2.46
8	2.13	3.10	2.57
10	2.18	3.28	2.72
12	2.29	3.44	2.78
14	2.33	3.66	2.85

Observation: Trained with chunk = **10**, evaluated with smaller *k* → **better accuracy** (easier task than training). Larger *k* (12, 14) extrapolates beyond training horizon → **accuracy degrades**. Higher *k* still gives better **throughput** (more actions / inference).

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Past Delivery vs Current Model — *extended multi-gesture*

Mean MAE on `extended_gesture_mimic` held-out test set

Actions / step (k)	Past delivery `smolvla-multigesture-val` (ckpt 010000)	Current model `smolvla-gesture3-trainall-evaltest` (ckpt 030000)
1	—	2.57
2	—	2.56
4	—	2.76
6	—	2.93
8	—	3.10
10	5.39	3.28
12	—	3.44
14	—	3.66

Observation: Current model (combined training, ckpt 030000) cuts MAE on `extended_gesture_mimic` from **5.39 → 3.28** at $k = 10$ — a **~39% reduction** over the past delivery.

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Possible Next attempt: Train with Larger Chunks

Why push the training chunk beyond 10?

Current state (chunk = 10)

- Eval with $k < 10$ → easy, better accuracy
- Eval with $k > 10$ → unseen horizon, accuracy drops
- Throughput is capped by **k = 10** for safe accuracy

Hypothesis

- Train with **larger chunk** (e.g. 16 / 20)
- Then chunk = 10 at inference becomes an **easier** sub-task
- Expect: **same-or-better accuracy at k = 10** with **higher throughput headroom**

Throughput math (Strix, bf16)

Quantity	Value
Inference rate (model)	**2.3 inf / sec**
Actions per inference (k)	**10**
Effective action rate	**23 actions / sec**

Next experiments

- Retrain with chunk ∈ {16, 20}
- Re-run the k-sweep on all 3 held-out test sets
- Verify: at $k = 10$, accuracy ≥ current chunk-10 model
- Then push k higher to trade marginal accuracy for throughput

Goal: decouple **training horizon** from **inference chunk** — train hard, infer fast.

Take-Home Message

- SmolVLA trained on **combined** (multi-gesture + extended + action-per-episode) data, chunk = 10.
- Held-out evaluation across all 3 sources; **k-sweep** shows the expected pattern: **smaller k → better accuracy**, **larger k → faster but less accurate**.
- **Next step:** train with a **larger chunk** so that $k = 10$ at inference is well inside the training distribution → preserve accuracy while unlocking higher action throughput on Strix (currently 23 actions/sec @ k = 10, bf16).

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